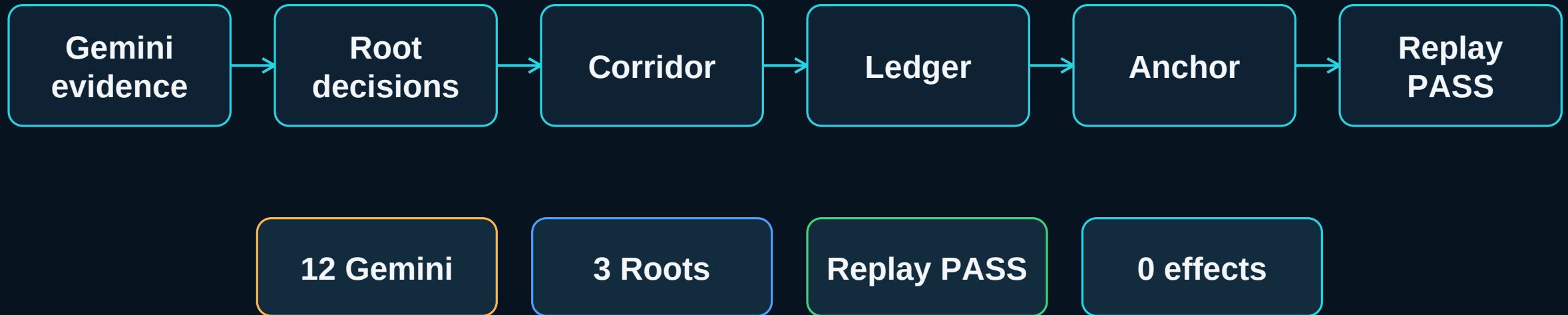


Cover - central thesis

HEDGEHOG OS separates LLM reasoning from authority, then preserves the accepted decision path as a deterministic, cryptographically anchored, replayable evidence chain.



Executive proof in 90 seconds

12
real Gemini actors

3
sovereign Roots

1
**deterministic
transaction**

19
Ledger artifacts

29
dependency edges

3
Root finals

9
Crypto source files

11
**critical package
files**

1
committed Anchor

19
Replay rows

0
real-world effects

The user's mock travel task and accepted result

PAR -> LIM
2026-08-12
Preference A

Selected Offer A
offer:mock_airline_al:PAR-LIM:001
tri_airline_purchase:PAR-LIM:2026-08-12:client_001

MOCK / PROOF-LEVEL EVIDENCE
NOT A REAL TICKET - NOT A REAL BOOKING - NOT A REAL
PAYMENT

Complete architecture from user task to anchored Replay

REASONING**AUTHORITY****DETERMINISTIC****EVIDENCE + VERIFY**

The twelve real Gemini actors and bounded topology

Tri-Party Orchestrator

Semantic Architect

Client Purchase Intent

Client Privacy

Airline Offer Policy

Fare Rules Cell

Seat & Baggage Cell

Ticketing Policy

Bank Payment Policy

Idempotency Risk Cell

Payment Status Explainer

Cross-Root Consistency

5 causal / 7 generic / all advisory

Orchestration and Client-side actor evidence

1. tri_party_airline_orchestrator_IIm

Interpret the whole tri-party goal and propose the bounded semantic route that precedes BSEP creation

The transaction should be routed through a bounded tri-party semantic program and a BSEP membrane before downstream architecture work.

PASS / accepted true / ADVISORY ONLY

2. tri_party_airline_semantic_architect_IIm

Propose a semantic actor topology and validation obligations from BSEP-derived bounded context
A bounded actor topology and validation plan can organize the client, airline, bank, vertical-cell, and cross-root reviews.

PASS / accepted true / ADVISORY ONLY

3. client_purchase_intent_reviewer_IIm

Review which hard-compatible offer best matches the client's soft travel preferences

Offer `offer:mock\_airline\_al:PAR-LIM:001` best matches the lower-price and window-seat preferences while the compared offers remain hard-compatible.

PASS / accepted true / ADVISORY ONLY

4. client_profile_privacy_reviewer_IIm

Review how passenger and payment references remain opaque and how private fields stay bounded
Passenger and payment references should remain opaque, with access controlled through bounded privacy mechanisms rather than raw-field exposure.

PASS / accepted true / ADVISORY ONLY

Airline-side actor evidence

5. airline_offer_policy_reviewer_llm

Review the proposed Airline offer under bounded policy evidence

Offer `offer:mock\_airline\_al:PAR-LIM:001` was reviewed as acceptable advisory policy evidence.
PASS / accepted true / ADVISORY ONLY

6. airline_fare_rules_vertical_cell_llm

Perform a bounded child review of fare-rule considerations for the proposed offer

Offer `offer:mock\_airline\_al:PAR-LIM:001` passed the bounded fare-rules advisory review.
PASS / accepted true / ADVISORY ONLY

7. airline_seat_baggage_vertical_cell_llm

Perform a bounded child review of seat and baggage considerations for the proposed offer

Offer `offer:mock\_airline\_al:PAR-LIM:001` passed the bounded seat/baggage advisory review.
PASS / accepted true / ADVISORY ONLY

8. airline_ticketing_policy_reviewer_llm

Review the preconditions for mock ticket evidence
Mock ticket evidence requires a confirmed hold, successful mock payment authorization, and explicit client approval

PASS / accepted true / ADVISORY ONLY

Bank-side and cross-root actor evidence

9. bank_payment_policy_reviewer_llm

Review the bounded mock payment-authorization evidence

The bounded fields describe a mock authorization proposal and protected payment reference, not settlement or an external bank action.

PASS / accepted true / ADVISORY ONLY

10. bank_idempotency_risk_vertical_cell_llm

Review duplicate-payment, idempotency, expiry, TTL, amount, and merchant mismatch risk
The bounded mock authorization evidence passed the idempotency and mismatch-pressure review.

PASS / accepted true / ADVISORY ONLY

11. bank_payment_status_explainer_llm

Explain the evidence distinction between mock authorization and settlement

A payment-status receipt is evidence of a mock state

PASS / accepted true / ADVISORY ONLY

12. tri_party_evidence_consistency_reviewer_llm

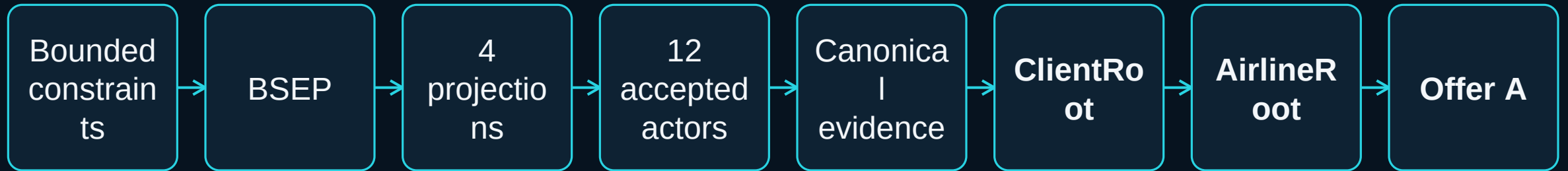
Review consistency across client, airline, and bank evidence

Offer `offer:mock\_airline\_al:PAR-LIM:001` remained coherent across the reviewed evidence.

PASS / accepted true / ADVISORY ONLY

Why the deterministic system selected Offer A

No provider override - no silent fallback - provider output is not truth or authority



offer:mock_airline_al:PAR-LIM:001

Three sovereign Roots and authority separation

ClientRoot
client constraints
purchase intent
client acceptance
ClientRoot final

AirlineRoot
offer resolution
hold
ticket issue intent
mock ticket receipt
AirlineRoot final

BankRoot
payment authorization
boundary
payment evidence
BankRoot final

Gemini != Root Reviewer != Root Crypto != Root Replay != Root

Five-phase Ticket/Purchase Corridor and mock evidence card



Hold packet	airline_hold_commit_packet:semantic_causal:001
Client purchase intent	client_purchase_intent:client_001:001
Bank authorization ref	bank_payment_authorization_ref:mock_bank_a:001
Ticket issue intent	airline_ticket_issue_commit_packet:mock_airline_al:001
Mock ticket receipt	mock_ticket_receipt:mock_airline_al:001
Mock purchase receipt	mock_purchase_receipt:client_001:001
Root finals	client_root_final:001 airline_root_final:001 bank_root_final:001

MOCK EVIDENCE - NO REAL TICKET / BOOKING / PAYMENT

Transaction Artifact Ledger - 19 / 29 / 3



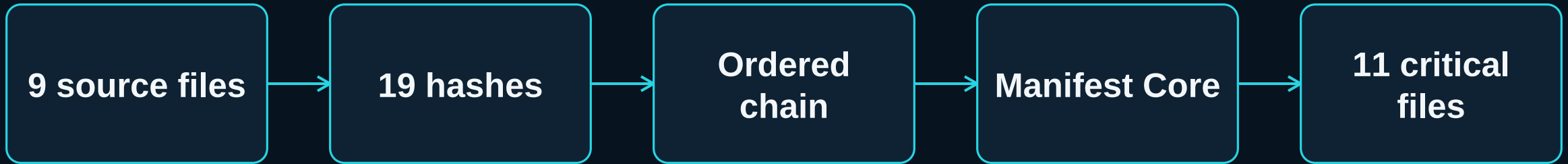
19 artifacts
29 dependency edges
3 Root finals

Blue: ClientRoot
Teal: AirlineRoot
Violet: BankRoot
Amber: advisory/evidence

All dependencies point backward.

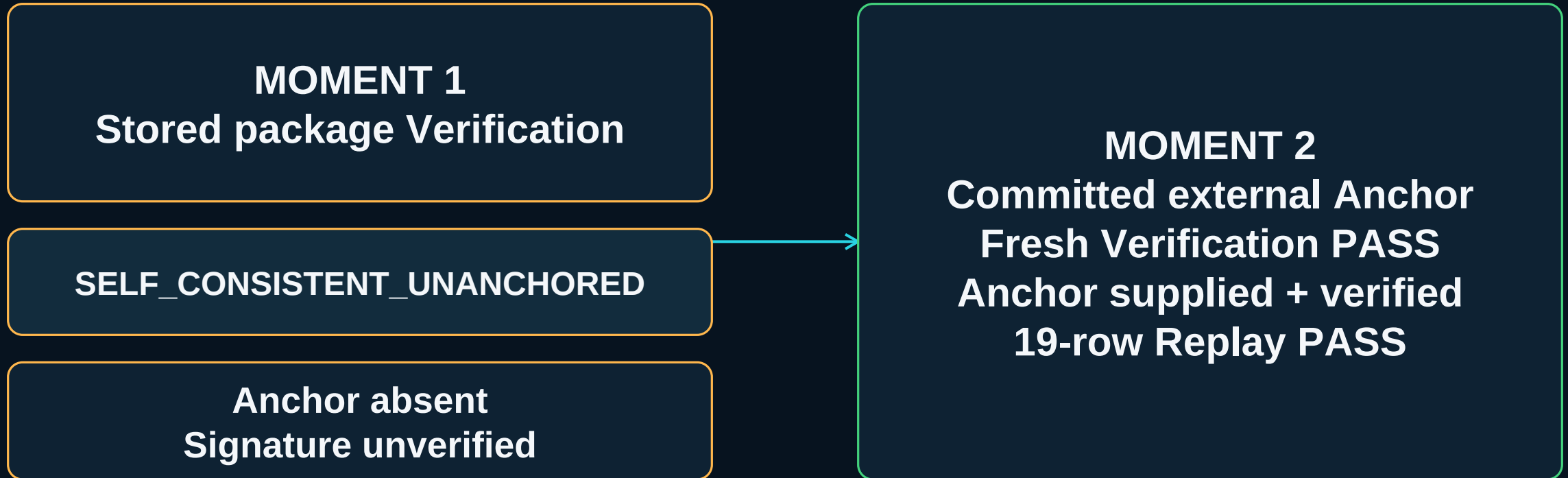
Crypto generation - 9 / 11 and SELF_CONSISTENT_UNANCHORED

Generation status: SELF_CONSISTENT_UNANCHORED



No external Anchor at generation. No anchored PASS claimed.
UNSIGNED_PLACEHOLDER / signature verified false.
Integrity is not semantic truth.

Committed external Anchor and 19-row Replay PASS



Replay provider / network / Gemini: 0 / 0 / 0 Package bytes unchanged: true

First honest FAIL_CLOSED run and distinct successful recovery

FIRST OFFICIAL PACKAGE
12 real Gemini calls completed
Ledger compatibility false positive
FAIL_CLOSED
No Anchor / no Replay
Preserved / not promoted



DISTINCT RECOVERY
Repair commit 38ad0b0
411 focused tests
1230 compatibility tests
Separate package
Generation PASS
Anchored Replay PASS

What was proved, non-claims, and evidence entry points

WHAT WAS PROVED

Bounded real LLM reasoning
Root authority separation
Deterministic Corridor
Causal Ledger
Byte-integrity Crypto Seal
Committed Anchor
Zero-rerun Replay

WHAT WAS NOT PROVED

No real ticket / booking / payment
No production integration
No PKI / signer authentication
No Root Attestation
Crypto is not semantic truth
Replay is not effect authorization

Evidence entry points:

[docs/airline_all_real_full_stack_human_story_v01.md](#) | [docs/airline_all_real_full_stack_crypto_anchor_v01.json](#)

[docs/airline_all_real_full_stack_replay_report_v01.json](#) | [docs/audit_reports/auditor_airline_all_real_full_stack_v01_generation_anchor_publication.log](#)

[docs/audit_reports/auditor_airline_all_real_full_stack_v01_anchored_replay.log](#) | [docs/showcase/airline_all_real_full_stack_v01/claim_evidence_matrix_v01.json](#)